EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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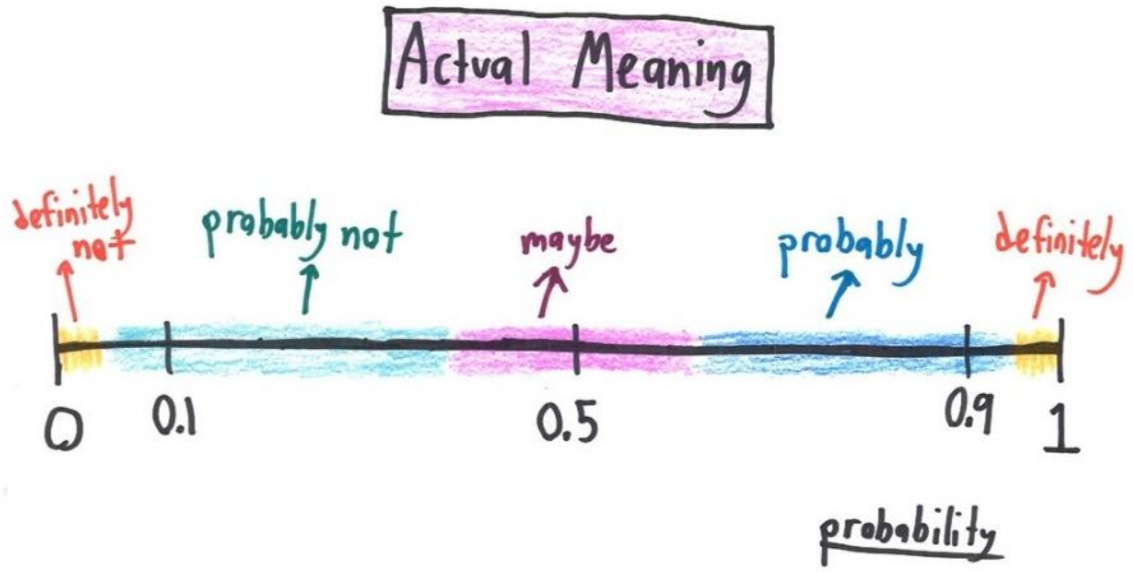
Week 4 - Lecture No 06 – 10th September 2025



Announcements!

1. Lectures up to No 05 posted
2. Quiz 1 solution and grades posted
3. Quiz No 02 – next week
4. An added lecture will be posted in a asynchronous mode in lieu of September 08, 2025 class

Agenda for today

1. Complete Unit 2 - Conditional Probability and Bayes' Rule

Bayes' Rule & inference

❖ Thomas Bayes, Presbyterian minister (c. 1701-1765)

❖ "Bayes' Theorem" published posthumously

❖ Systematic approach for incorporating new evidence

➢ Bayesian Inference

- Initial beliefs $P(A_i)$ on all causes of an observed event B.
- Model of the world under each A_i: $P(B \mid A_i)$

$$A_i \xrightarrow[P(B \mid A_i)]{\textbf{Model}} B$$

- Draw conclusions about causes

$$B \xrightarrow[P(A_i \mid B)]{\textbf{inference}} A_i$$

1) $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

2) $P(B|A) = \dfrac{P(A \cap B)}{P(A)}$

3) $P(B|A) = \dfrac{P(A|B) \cdot P(B)}{P(A)}$

Example: Baye's Rule

A new test (T) has been developed to determine whether a given student is overstressed (S). This test is 95% accurate if the student is not overstressed, but only 85% accurate if the student is in fact overstressed. It is known that 99.5% of all students are overstressed. *Given that a particular student tests negative for overstress, what is the probability that the test results are correct, and that this student is not overstressed?*

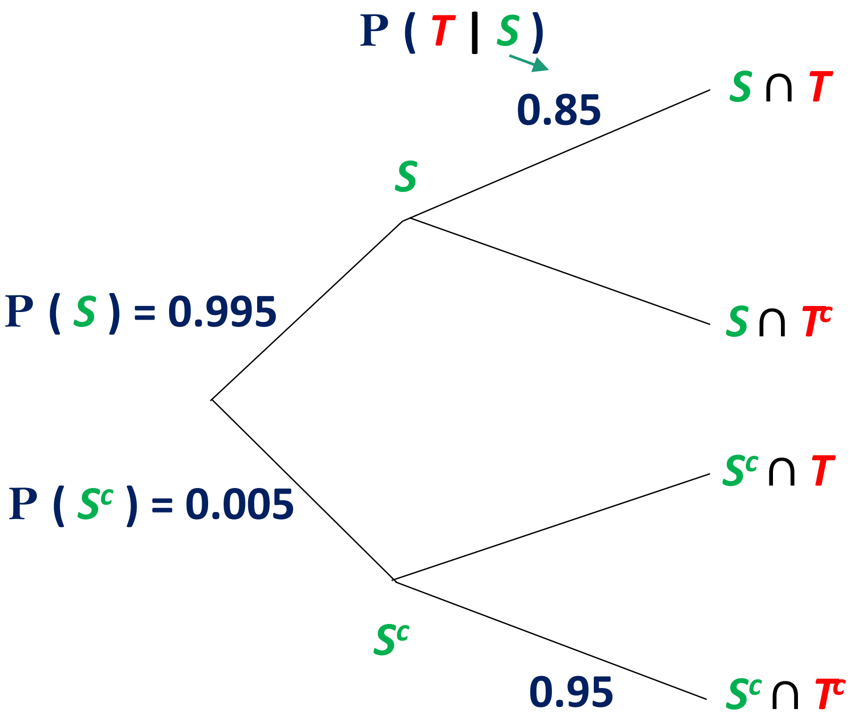
S = Student is over-stressed, S^c = Student is NOT over-stressed

T = Test positive, T^c = Test negative

$$\textbf{Find } P(S^c|T^c) = \frac{P(T^c|S^c) \times P(S^c)}{P(T^c)}$$

$$= \frac{0.95 \times 0.005}{0.95 \times 0.005 + 0.15 \times 0.995}$$

$$= 0.0308$$



1) $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

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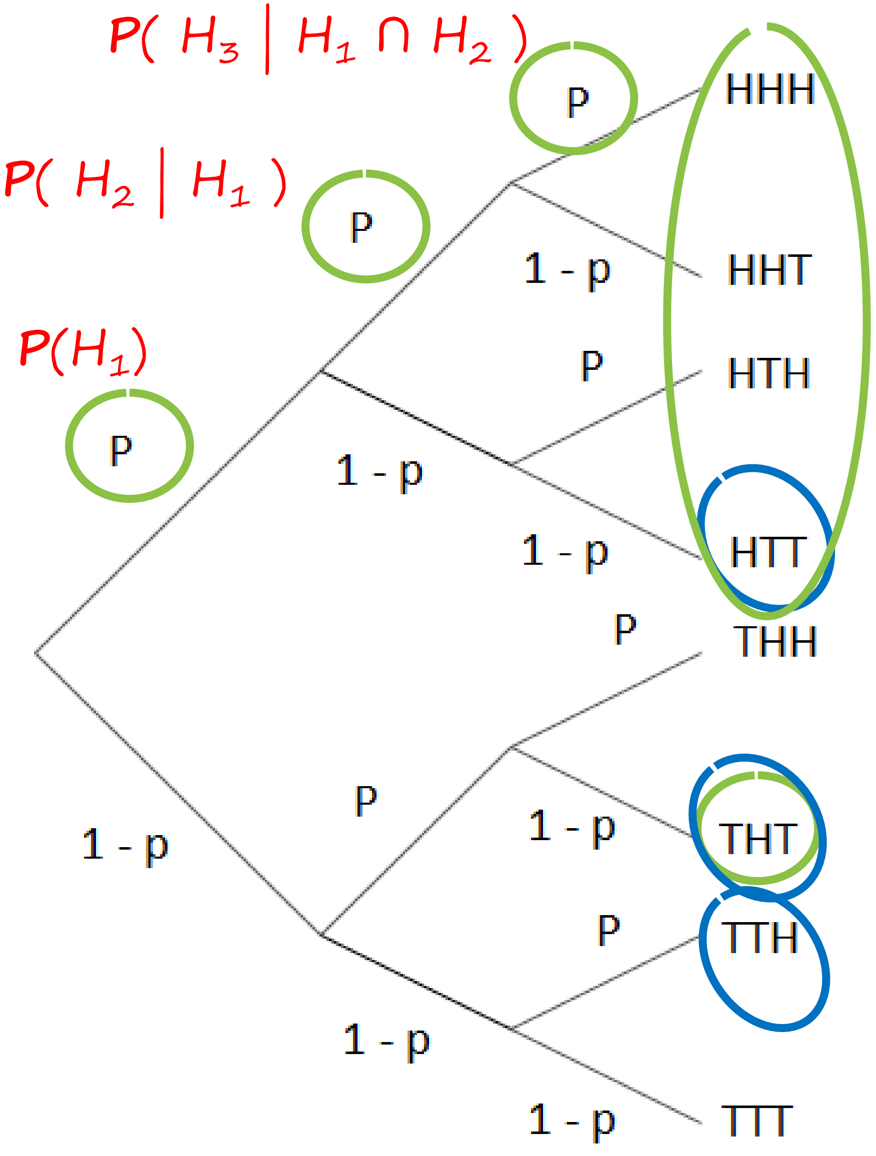
3) $P(B|A) = \dfrac{P(A|B) \cdot P(B)}{P(A)}$

A model based on conditional Probabilities

Let's consider 3 tosses of biased coin: $\mathbf{P}(H) = p$; $\mathbf{P}(T) = 1 - p$



1. Multiplication Rule: $\mathbf{P}(T\,H\,T) = (1-p)\; p\; (1-p)$
2. Total Probability: $\mathbf{P}(\text{1 Head}) = 3\,p\,(1-p)^2$
3. Bayes Rule:

$$\mathbf{P}(H_1 \mid \text{1 Head}) = \frac{P(H_1 \cap \text{1 head})}{P(\text{1 head})}$$

$$= \frac{p(1-p)^2}{3\,p(1-p)^2} = \frac{1}{3}$$

Few Comments

$P(H_2 \mid H_1) = p = P(H_2 \mid T_1)$

Any deductions?

$$\mathbf{P}(H_2) = P(H_1)\,P(H_2 \mid H_1) + P(T_1)\,P(H_2 \mid T_1)$$
$$= p \cdot p + (1-p) \cdot p$$
$$= p$$

So, $\mathbf{P}(H_2 \mid H_1) = p = \mathbf{P}(H_2 \mid T_1)$

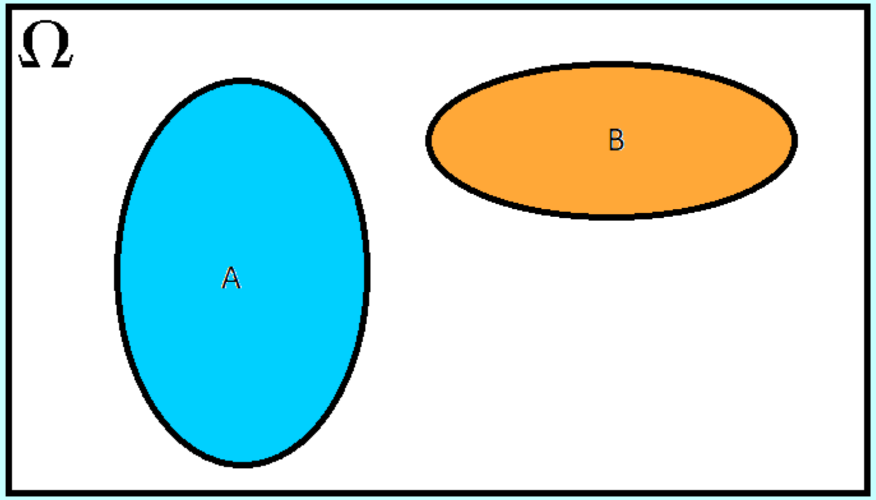
Independence of Two Events

- Intuitive "definition" : **P**(B|A) = **P**(B)
  - Occurrence of A provides no new information about B

$$P(A \cap B) = P(A) * P(B|A)$$
$$= P(A) * P(B)$$

Definition of Independence: $\mathbf{P}(A \cap B) = \mathbf{P}(A) * \mathbf{P}(B)$

- Symmetric with respect to A and B
- It also implies **P**(A|B) = **P**(A)
- Also applies even when **P**(A) = 0



P(A) > 0 P(B) > 0

Independent?

$$P(A \cap B) = 0$$
$$P(A) * P(B) > 0$$

<u>**Note: Being Independent & being disjoint are different things.**</u>

Conclusion?

Two events are independent if occurrence of one tells you nothing about the other.
For example:

A : Flip of a coin
B : It's raining

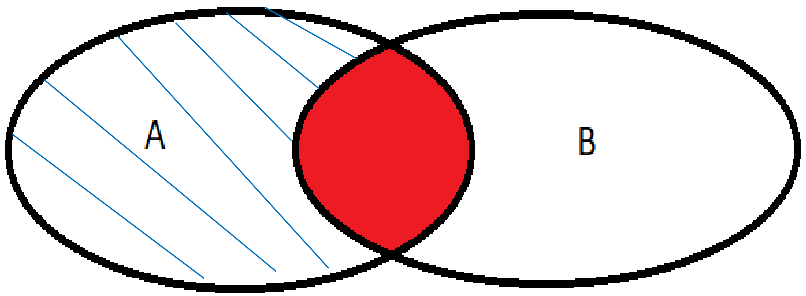
The above events are physically distinct and non-interacting

Note: In some cases it is less obvious

Independence of Event Complements

Definition of Independence: $P(A \cap B) = P(A) . P(B)$

- If A and B are independent, then A and B^c are independent.
  - Intuitive Argument



  - **Formal proof**

$$A = (A \cap B) \cup (A \cap B^c)$$

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

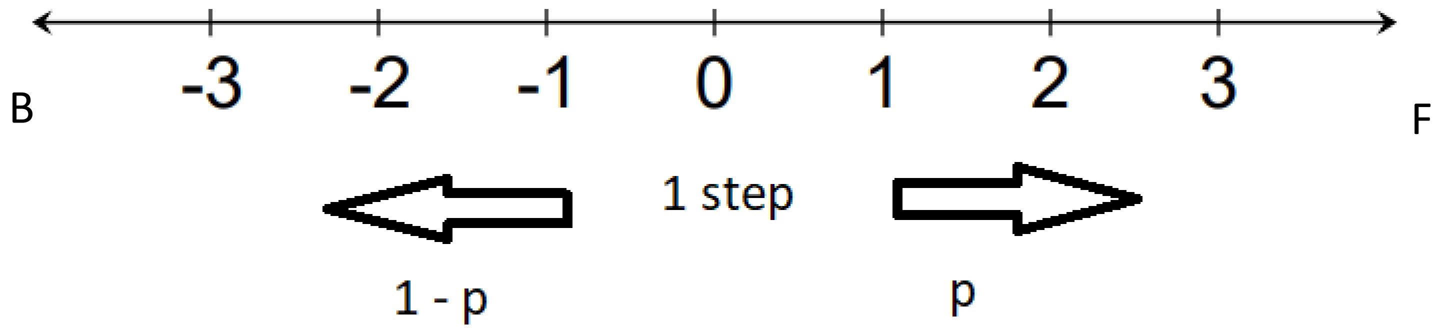
$$= P(A)P(B) + P(A \cap Bc)$$

$$P(A \cap B^c) = P(A) - P(A)P(B)$$

$$= P(A)(1 - P(B))$$

$$= P(A)P(B^c)$$

Let's consider an example of a Random Walker



a) **P**(After 2 steps the walker is at the origin) ? Note: Let's call this event A

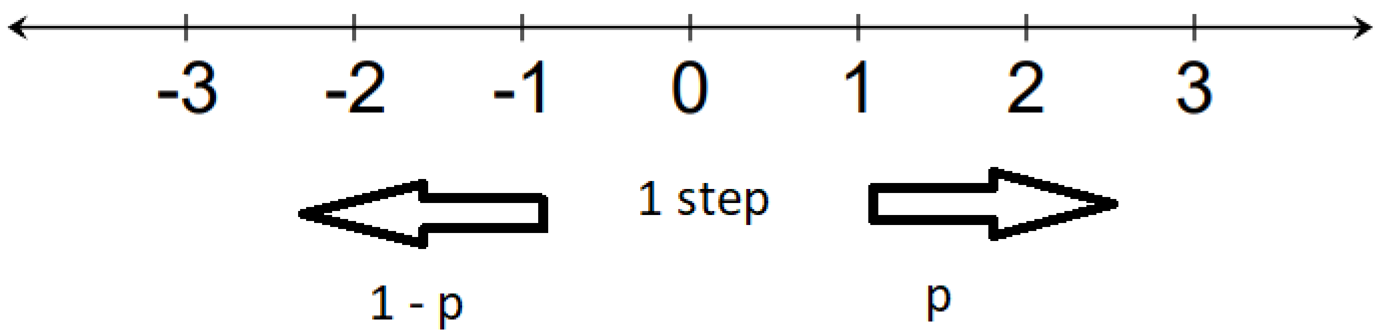
➡ $A = F_1B_2 \text{ or } B_1F_2$

➡ $P(A) = P(\{ (F_1B_2), (B_1F_2) \})$

➡ $= P(F_1B_2) + P(B_1F_2)$

➡ $= 2p(1-p)$

Let’s consider an example of a Random Walker



b) **P**(After 3 steps, the walker is at +1) ? Note: Let’s call this event B

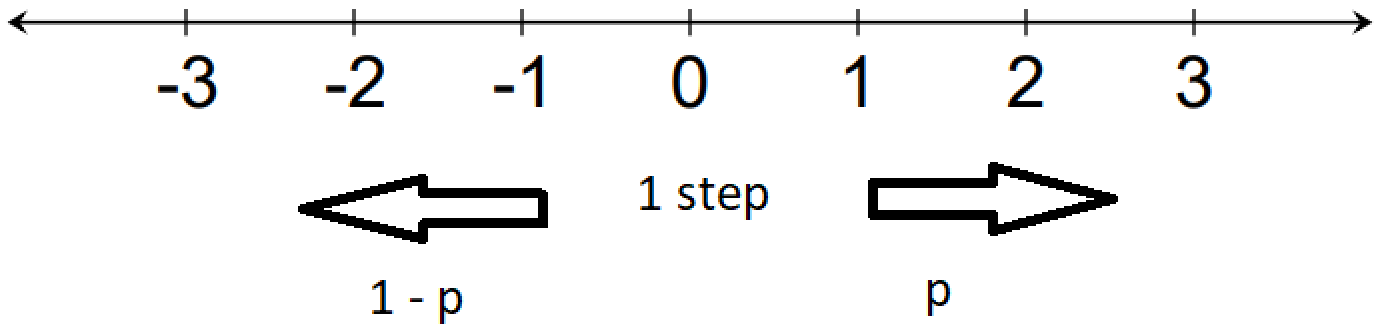
➡ $B = F_1F_2B_3 \text{ or } F_1B_2F_3 \text{ or } B_1F_2F_3$

➡ $P(B) = P(\{ (F_1F_2B_3), (F_1B_2F_3), (B_1F_2F_3)\})$

➡ $= 3\, p\, p\, (1-p)$

➡ $= 3\, p^2 (1-p)$

Let's consider an example of a Random Walker



c) **P**(1st step is backwards | B) ?

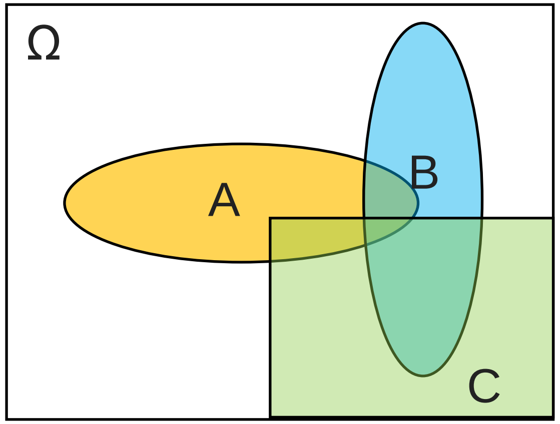
$$B = F_1F_2B_3 \text{ or } F_1B_2F_3 \text{ or } B_1F_2F_3$$

$$P(B) = 3\,p^2(1-p)$$

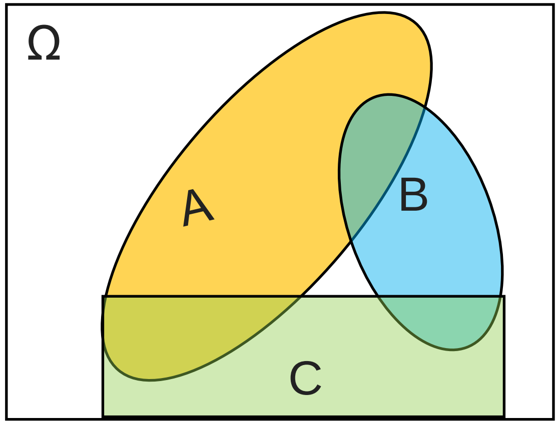
$$\Rightarrow \frac{P(B_1 \cap B)}{P(B)} = \frac{P(B_1F_2F_3)}{3\,p^2(1-p)} = \frac{p^2(1-p)}{3\,p^2(1-p)} = \frac{1}{3}$$

Conditional Independence

- Conditional independence, given an event C,
  is defined as independence under the conditional probability law **P**(· | C)



Assume A and B are independent



- If we are told that C occurred,
  are A and B independent? *No*

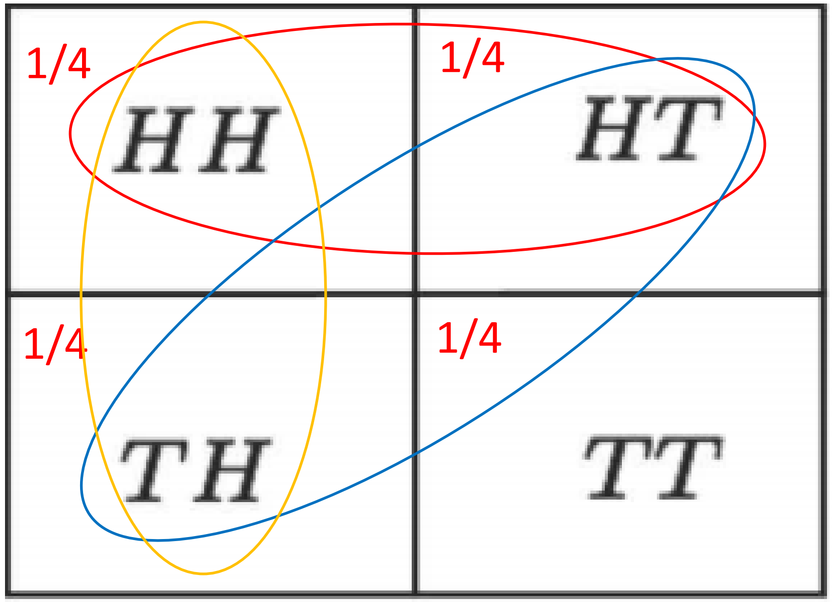
$$P(A \cap B \mid C) = P(A \mid C) * P(B \mid C)$$

An Example

- Two independent fair coin tosses
  - H_1 : First toss is H
  - H_2 : Second toss is H

$$P(H_1) = P(H_2) = ½$$

- C : the two tosses had a different result



$\Rightarrow P(H_1 \cap H_2) = P(HH) = ¼$

$\Rightarrow \quad = P(H_1)\, P(H_2) = \frac{1}{4}$

H_1 & H_2 are independent events

$P(H_1 \cap H_2 \mid C) = 0$

$(H_1 \cap H_2 \mid C)$ are not independent

$\Rightarrow \quad \neq P(H_1 \mid C)\, P(H_2 \mid C) = ¼$

Puzzle – The King's Sibling

- The king comes from a family of two children. What is the probability that his sibling is female?

Assumptions
- Boys have precedence – King is always a boy
- $\mathbf{P}$ (Boy) = $\mathbf{P}$ (Girl) = ½, independent

$B_1\ B_2$ ¼	$G_1\ B_2$ ¼
$B_1\ G_2$ ¼	$G_1\ G_2$ ¼

So, What is the answer?

King's sibling has equal chance of being a boy or a girl, therefore, $\mathbf{P}$ (sibling is a girl) = ½ X

➡ It is given that we do have a King

$B_1\ B_2$ 1/3	$G_1\ B_2$ 1/3
$B_1\ G_2$ 1/3	$\cancel{G_1\ G_2}$

➡ is this a correct answer? NO

➡ Therefore, $\mathbf{P}$ (sibling is a girl) = 2/3